

omicstra TNBC-92

omicstra · TNBC-92 v3 - calculation provenance map

the technical companion to `tnbc92_methods.md` (which teaches the *concepts*). this doc records **where every number comes from**: the script + function that computes each metric, the config that defines each run, and the artifact each result lands in. it exists because `scripts/` is gitignored (QA workspace, per project convention) - so this tracked doc is the durable lineage record for review and reproduction.

grid: `runs/tnbc-92_v3/` · **niche-join**: `data/embeddings/niches_v3/` (sha-locked `gpath2vec 13985cbd...`) · **split**: patient-stratified 85/15, seed 42, shared across runs via `split.json`.

stage 1 · niche-join (per-niche feature table)

what	where
build script	<code>scripts/build_niche_join.py</code>
key functions	<code>process_subarray()</code> (per-niche row build), <code>parse_niche_id()</code> (subarray:: <code>spot</code> key), <code>niche_mean_pool()</code>
inputs	<code>data/embeddings/{virchow2_niche, virchow2_cell, novae_niche_full}/</code> , <code>gpath2vec</code> parquet, <code>biological_signals/</code> <code>{mc_labels, mc_weights, tls_scores, morphology_labels}.tsv</code> , <code>clinical/Clinical.RDS</code>
output	<code>data/embeddings/niches_v3/{subarray}.parquet</code> + <code>manifest.json</code> (records <code>gpath2vec</code> sha256, git commit, <code>mc_label</code> coverage)

what	where
	python scripts/build_niche_join.py --gpath2vec-parquet data/embeddings/gpath2vec/
invocation	fisher_madmean_low_dim512_e5_s1234/ fisher_madmean_low_cluster_embeddings.parquet --out-dir data/embeddings/niches_v3

stage 2 · alignment runs

run	script	config	output artifacts
R1-R4, R6	scripts/align.py	configs/v3/ {R1,R2,R3,R4,R6} _v3.json	runs/tnbc-92_v3/{run}_v3/ {embeddings_test.parquet, metrics_h1_raw.json, run_config.json, split.json, training_log.csv, checkpoint.pt}
R5 (AnInfoNCE)	scripts/align.py	configs/v3/ R5_v3.json	same; AnInfoNCE wiring = Config.anisotropic, aninfonce_loss(), LateFusion(anisotropic=)
B1/B2/B3	scripts/ align_classical.py	--baseline cca\ procrustes\ unaligned	same artifact schema, closed-form (no checkpoint/training_log)
B4	scripts/ align_b4.py	--reference- split runs/ tnbc-92_v3/ R1_v3/split.json	random-init MLP, run_config.json records training_steps: 0
model classes	align.py: LateFusion, CrossAttnFusion (R4), MLPBlock		attention_weights column written by CrossAttnFusion only (R4)
losses	align.py: infonce_loss, supcon_loss, barlow_loss, aninfonce_loss		

stage 3 · metrics - function -> algorithm -> artifact

metric	function (file)	algorithm	lands
AUC	retrieval_metrics() (align.py)	sklearn roc_auc_score on matched-diagonal vs sampled- mismatched cosines	runs/
R@K, MRR, median rank, alignment gap	retrieval_metrics() (align.py)	argsort ranking of self in cosine matrix	same
CKA	cka_linear() (align.py)	linear-kernel HSIC ratio on centered z_he vs z_st	same
H2-A archetype ARI + silhouette	cluster_quality() (eval.py)	sklearn KMeans(9 or 14, n_init=10, seed42) -> adjusted_rand_score + silhouette_score	runs/
H2-A mc_megacluster ARI	cluster_quality() (scripts/ _scratch/eval_h2_mc_coherence.py)	KMeans(14) vs mc_megacluster, z_he	runs/
H2-A kNN purity	knn_purity() (scripts/_scratch/ eval_h2_mc_diagnostics.py)	sklearn NearestNeighbors (cosine via L2+euclidean), fraction of k-NN sharing mc	runs/ knn_p
H2-A linear probe (R4 rescue)	linear_probe() (eval_h2_mc_diagnostics.py)	sklearn LogisticRegression (lbfgs, C=1, 14-class), patient-stratified 11/3, StandardScaler	.../m
H2-A rank- matched KMeans	run_rank_matched_diagnostic() (eval_h2_mc_diagnostics.py)	project z_he to top-3 PCs (SVD), recompute ARI	.../m
	welch_z() (scripts/_scratch/ eval_h2_compartment_welch.py)	unpooled-variance z on per-	runs/ compa

metric	function (file)	algorithm	lands
H2-B.1 compartment Welch z		compartment matched-pair cosine aggregates	compa eval.
H2-B.2 TLS- signature z_TLS	univariate_cca_proj() + perm loop (scripts/_scratch/ eval_h2_tls_signature.py)	lstsq canonical direction on z_he vs continuous tls score, cross-patient 11/3, 500-perm null, bh_fdr()	runs/ tls_s
H2-C bio/patient ratio	eval_alignment_biology.py (perm z- tests)	matched-null permutation z on same-vs-diff-label cosine deltas for TIME/MC_global/ MC_tumor and patient_id	runs/ biolo
H3 pathway CCA z_A (AUCell arm)	univariate_cca() + perms (scripts/ eval_h3_pathway_cca.py, _perms.py)	lstsq+pearsonr, option-A cross- patient 11/3, perm null, BH-FDR	runs/ {per_
H3 pathway CCA z_A (gpath2vec arm, headline)	scripts/ eval_h3_pathway_cca_gpath2vec_v2.py	niche-cosine to embedded Reactome subtree, same option-A protocol; sha-locked embeddings	runs/ pathw {per_ json
BH-FDR	bh_fdr() (in the tls / perms scripts)	Benjamini- Hochberg over each declared test family	inline column

the scorecard + ROC figures (notebook §11, §12)

figure	reads from	built by
§11 scorecard (5 panels)	the 5 eval sources above (H1 json, mc_coherence, tls_signature_cca, biology.parquet, pathway_cca_gpath2vec_v3)	notebooks/final/ 05_summary_umaps_v3.ipynb §11 cell (reproducible via scripts/_scratch/ build_v3_summary_notebook.py)

figure	reads from	built by
§12 H1 ROC curves	each run's embeddings_test.parquet (recompute matched/ mismatched cosines) + AUC from metrics_h1_raw.json	§12 cell, same build script

CLI conventions (all eval scripts)

every eval script takes `--runs-dir runs/tnbc-92_v3 --niches-dir data/embeddings/niches_v3 --runs R1_v3,R2_v3,...` (click-based, per memory feedback_cli_use_click). v1 defaults are preserved when args are omitted, so the same scripts reproduce the v1 grid.

reproduce a number end-to-end (worked example: R4 AUC 0.859)

1. `data/embeddings/niches_v3/` exists (stage 1) ->
2. `python scripts/align.py --config configs/v3/R4_v3.json --runs-root runs/tnbc-92_v3` writes `runs/tnbc-92_v3/R4_v3/metrics_h1_raw.json` ->
3. `["auc"]` field = 0.8591, computed by `retrieval_metrics()` via `roc_auc_score` on matched vs sampled-mismatched cosines of `embeddings_test.parquet`'s `z_he/z_st`. the ROC curve for it is notebook §12.

see `tnbc92_methods.md` for what these metrics *mean* and `tnbc92_results_summary.md` for the *values*.